Exemptions

10.2.10 (1) Sections 10.2.1 to 10.2.5 do not apply to placer mines, sand and gravel pits and quarries unless required by the chief permitting officer.

(2) Sections 10.5.7, 10.5.8 and 10.5.9 do not apply to a TSF that
- (a) was granted the initial permit before July 20, 2016,
- (b) is not active, and
- (c) has no adverse material change after the date this subsection comes into force.

(3) Sections 10.5.2, 10.5.4, 10.5.5, 10.5.7, 10.5.8 and 10.5.9 do not apply to a category 3 dam that
- (a) was granted the initial permit before the date this subsection comes into force,
- (b) is low consequence as determined in accordance with section 10.5.6, and
- (c) does not contain tailings.

Exemptions and Obligations for Category 1A, Category 1B and Category 2 Dams

10.2.11 (1) Sections 10.4.1 to 10.6.13 do not apply to category 1A dams, category 1B dams and category 2 dams.

(2) A dam is a category 1A dam if
- (a) the dam is at a placer mining site, a sand and gravel pit, a rock quarry or an industrial mineral mine, and
- (b) the manager assesses that the following apply:
  - (i) the dam meets all the requirements in Table 10-1;
  - (ii) the dam contains no tailings;
  - (iii) there is no identifiable human population at risk of injury in the event of a breach of the dam other than through unforeseen misadventure;
  - (iv) there is no potential for human loss of life in the event of a breach of the dam;
  - (v) there are no seasonally or permanently occupied buildings or infrastructure within 500 m downstream of the dam;
  - (vi) the dam does not contain toxic or deleterious substances, excluding suspended sediment.

(3) A dam is a category 1B dam if
- (a) the dam is at a metal or coal mine, and
- (b) a qualified professional assesses that the following apply:
  - (i) the dam meets all the requirements in Table 10-1;
  - (ii) the dam contains no tailings;

(iii) in the event of a breach of the dam, other than through unforeseen misadventure,
  (A) the effect of water released to receiving streams on aquatic or terrestrial habitat is short-term and reversible,
  (B) there is no identifiable human population at risk of injury, and
  (C) there is no potential for human loss of life;
(iv) there are no seasonally or permanently occupied buildings or infrastructure within 500 m downstream of the dam.

(4) A dam is a category 2 dam if a professional engineer
  (a) assesses that the dam meets all the requirements in Table 10-2, and
  (b) submits the assessment to the chief inspector.

(5) The assessments referred to in subsections (2) (b), (3) (b) and (4) (a) must be made available to an inspector on request.

(6) The manager must ensure that all category 1A dams, category 1B dams and category 2 dams are properly inspected, maintained and repaired in a manner that keeps the dams in good operating condition.

Table 10-1 Category 1A & Category 1B Dam Requirements

Dam Criteria	Required Value
Height* (metres) x Storage volume (metres^3)	<20,000 m^4
Maximum height*	2.5m
Minimum crest width	3m
Dam slopes	2H:1V or flatter
Maximum water level	0.5m below the crest

*as measured from the lowest point of the downstream toe of the dam to the crest of the dam.

Table 10-2 Category 2 Dam Requirements

Dam Criteria	Required Value
Maximum storage volume*	30,000 m^3
Maximum height**	2.5 m
Contents	Does not impound tailings
Dam slopes	2H:1V or flatter
Maximum water level	0.5 m below crest or as determined by P.Eng.
Consequences of potential failure scenarios	Low***

*as calculated based on the maximum water level.
**as measured from the lowest point of the downstream toe of the dam to the crest of the dam.
***as determined under section 10.5.6

Filing and Reporting

Publication

10.3.1 When required by an inspector, notice of filing an application under section 10 (1) of the *Mines Act* must be published, by the person filing it, in the Gazette and in local newspapers.

Written Response

10.3.2 If a notice of filing has been published under section 10.3.1 of this code, a person affected by, or interested in, the application has 30 days after the last date on which the notice was published to view the application and make written representations to the chief permitting officer.

Mine Plan, Reclamation Plan and Closure Plan

10.3.3 (1) Unless otherwise stated in the permit, the manager must ensure that the mine plan, reclamation plan and closure plan are

(a) developed and provided to the chief inspector on commencement of operations, and

(b) updated and provided to the chief inspector at least every 5 years.

(2) The manager must ensure that the reclamation plan outlines the progressive reclamation activities planned for the 5 years following the date on which the mine plan is updated in accordance with subsection (1) (b).

(3) The manager must

(a) ensure that reasonable efforts are made to engage with each affected First Nation in order for each First Nation to identify if it wants to receive the mine plan, reclamation plan or closure plan described in subsection (1), and

(b) provide to each First Nation a copy of each of the most recent plans identified by the First Nation under paragraph (a) of this subsection, in accordance with the timeframe applicable with respect to the chief inspector.

Additional Reporting Requirements

10.3.4 (1) In addition to other reporting requirements set out in this Part, the manager must

(a) annually provide to the chief inspector

(i) a description, in a form specified by the chief inspector, of the reclamation and environmental monitoring work referred to in section 10.2.2 (e), by March 31 of the following year,

(ii) a summary of outstanding TSF and dam orders issued by inspectors, including the scheduled completion dates, by March 31 of the following year, and

(iii) a report, respecting each affected First Nation, that lists

(A) the documents the First Nation identified under subsection (2) (a) of this section and sections 10.3.3 (3) (a), 10.4.3 (8) (a), 10.5.4 (3) (a), 10.6.2 (5) (a), 10.6.3 (3) (a), 10.6.4 (6) (a), 10.6.5 (6) (a), 10.6.7 (11) (a), 10.6.12 (8) (a) and 10.7.1 (3) (a), by March 31 of the following year, and

(B) the documents provided to the First Nation under subsection (2) (b) of this section and sections 10.3.3 (3) (b), 10.4.3 (8) (b), 10.5.4 (3) (b), 10.6.2 (5) (b), 10.6.3 (3) (b), 10.6.4 (6) (b), 10.6.5 (6) (b), 10.6.7 (11) (b), 10.6.12 (8) (b) and 10.7.1 (3) (b), by March 31 of the following year, and

(b) provide other reporting as directed by the chief inspector.

(2) The manager must

(a) ensure that reasonable efforts are made to engage with each affected First Nation in order for each First Nation to identify if it wants to receive any of the documents described in subsection (1) (a) (i) and (ii), and

(b) provide to each First Nation a copy of each of the most recent documents identified by the First Nation under paragraph (a) of this subsection, in accordance with the timeframe applicable with respect to the chief inspector.

TSFs and Dams – Responsibilities

Engineer of Record

10.4.1 (1) The manager must ensure that a professional engineer, who has the technical expertise and experience commensurate with the complexity of the TSF or dam, is retained as the engineer of record for each TSF and dam.

(2) The manager must, within 72 hours of an engineer of record accepting the role, provide the chief inspector with the engineer of record's written acknowledgement that the engineer of record

(a) has the commensurate expertise and experience referred to in subsection (1), and

(b) is accepting the role of engineer of record.

(3) The manager must provide the chief inspector with the engineer of record's written notification that the engineer of record is departing from the role of engineer of record, within 72 hours of receiving the notification.

(4) A reference to the engineer of record in this Part includes the qualified professionals under the supervision of the engineer of record, except with respect to
(a) the references to engineer of record in this section and sections 10.4.4, 10.6.2 (3) (a) and (b), 10.6.4 (3) and (4), 10.6.6 (7) (b), 10.6.7 (7) (b) and 10.6.9 (2), and
(b) the second reference to engineer of record in sections 10.5.2 (2), 10.5.3 (2) and 10.5.4 (2) (a).

TSF Qualified Person and Dam Qualified Person

10.4.2 (1) The manager must
(a) designate a TSF qualified person for safe management of each TSF,
(b) designate a dam qualified person for safe management of each dam not associated with a TSF,
(c) provide to the chief inspector a written acknowledgement,
(i) signed by the manager, that each qualified person referred to in paragraph (a) or (b) has the technical expertise and experience commensurate with the complexity of the TSF or dam, as applicable, and
(ii) signed by the qualified person, confirming the person accepts the role of qualified person under paragraph (a) or (b), as applicable, and
(d) if the qualified person under paragraph (a) or (b) changes, notify the chief inspector within 72 hours.

(2) A qualified person referred to in subsection (1) (a) or (b) must, as a result of the person's knowledge, training and experience,
(a) be qualified to organize, supervise and perform duties related to the safe management of the TSF or dam, as applicable,
(b) be familiar with the provisions of the *Mines Act*, the code and the regulations that apply to the safe management of TSFs or dams, as applicable,
(c) be capable of identifying potential or actual danger to people or the environment as it relates to TSFs or dams, as applicable, and
(d) be the holder of a supervisor's certificate, as required under section 1.12.7 of this code.

Independent Tailings Review Board

10.4.3 (1) The manager of a mine with one or more tailings storage facilities must establish an Independent Tailings Review Board, unless exempted by the chief inspector.

(2) The manager must ensure that reasonable efforts are made to engage with affected First Nations regarding the establishment of the

Independent Tailings Review Board prior to requesting an exemption as set out in subsection (1).

(3) The manager must ensure that the composition and qualifications of the Independent Tailings Review Board is commensurate with the complexity of the TSF.

(4) The manager must provide to the chief inspector
  (a) a list of the Independent Tailings Review Board members and their qualifications, and
  (b) an update when the Board's membership changes.

(5) The manager must ensure that terms of reference for the Independent Tailings Review Board are developed, and updated, taking into consideration the complexity of the TSF.

(6) The manager must make an annual report of the activities of the Independent Tailings Review Board that describes the following:
  (a) a summary of the reviews conducted that year, including the number of meetings, topics discussed and attendees;
  (b) whether the work reviewed that year meets the Board's expectations of good practice;
  (c) any conditions that the Board is aware of that may compromise tailings storage facility integrity;
  (d) signed acknowledgement by the members of the Board, confirming that the report is a true and accurate representation of their reviews.

(7) The manager must provide to the chief inspector the annual report referred to in subsection (6) by March 31 of the following year.

(8) The manager must
  (a) ensure that reasonable efforts are made to engage with each affected First Nation in order for each First Nation to identify if it wants to receive any of the documents described in subsections (4) and (6), and
  (b) provide to each First Nation a copy of each of the most recent documents identified by the First Nation under paragraph (a) of this subsection, in accordance with the timeframe applicable with respect to the chief inspector.

Duty to Report Unresolved Safety Issues

10.4.4 (1) If the manager receives written notification from the engineer of record that a TSF or dam safety deficiency is not being addressed in an appropriate time period, the manager must, within 72 hours of receiving the written notification,
  (a) report the deficiency to the chief inspector, and
  (b) provide a copy of the report referred to in paragraph (a) to the engineer of record.

(2) If the manager does not provide the report to the chief inspector and the engineer of record in accordance with subsection (1), the engineer of record must, within 72 hours immediately following the expiration of 72-hour period set out in subsection (1), report the unaddressed TSF or dam safety deficiency to the chief inspector.

TSFs and Dams – Design

Design Responsibility

10.5.1 The manager must ensure that tailings storage facilities and dams are designed by, or under the direct supervision of, the engineer of record.

Site Characterization

10.5.2 (1) The manager must ensure that the engineer of record develops a site characterization for each TSF and dam, which supports the design of the TSF or dam, and includes the following:
- (a) climate, hydrology and climate change;
- (b) summary of environmental setting;
- (c) site surficial geology, geomorphology and geohazards;
- (d) bedrock geology, geotechnical conditions, hydrogeology and seismotectonic conditions;
- (e) representative plans and cross-sections of interpreted geological and geotechnical units, and groundwater conditions;
- (f) a signed and sealed assurance statement, in a form specified by the chief inspector.

(2) The manager must ensure that the site characterization for the TSF or dam is reviewed and updated by the engineer of record when the engineer of record considers it appropriate.

(3) The manager must ensure that, when the site characterization for the TSF or dam is developed or reviewed,
- (a) reasonable efforts are made to engage with affected First Nations, and
- (b) local Indigenous knowledge received under paragraph (a) is considered.

Design Report

10.5.3 (1) The manager must ensure that, for each TSF and dam, a design report is developed by the engineer of record, that includes
- (a) an analysis of the following, to support the design of the TSF or dam:
  - (i) the site characterization as set out under section 10.5.2;
  - (ii) tailings characterization and management, in the case of TSFs;

(iii) the consequences of potential failure scenarios as set out under section 10.5.6;
(iv) the risk assessment as set out under section 10.6.8;
(v) stability, deformation and other assessments, as considered appropriate by the engineer of record;
(vi) seepage and groundwater management;
(vii) water balance and water management;
(viii) closure,

(b) consideration of the following:
(i) mining or tailings processes;
(ii) environmental factors;
(iii) hydrological conditions and other conditions associated with climate change,

(c) a description of the proposed quantifiable performance objectives, and

(d) the design summary document developed or updated, as appropriate, under section 10.5.4 (1).

(2) The manager must ensure that the design report developed under subsection (1) is reviewed and updated by the engineer of record when the engineer of record considers it appropriate.

(3) The manager must ensure that, when the design report is developed or reviewed,
(a) reasonable efforts are made to engage with affected First Nations, and
(b) local Indigenous knowledge received under paragraph (a) is considered.

Design Summary Document

10.5.4 (1) The manager must ensure that the engineer of record develops a design summary document that summarizes the items listed in section 10.5.3 (1) (a) to (c) in the form of a table specified by the chief inspector.

(2) The manager must ensure the design summary document is
(a) reviewed and updated by the engineer of record when the engineer of record considers it appropriate, and
(b) provided to the chief inspector by March 31 of the year following the year it is updated.

(3) The manager must
(a) ensure that reasonable efforts are made to engage with each affected First Nation in order for each First Nation to identify if it wants to receive the design summary document as developed under subsection (1) or updated under subsection (2), whichever is the most recent, and

(b) provide a copy of the most recent document to each First Nation that identifies under paragraph (a) of this subsection, it wants to receive it, in accordance with the timeframe applicable with respect to the chief inspector.

Failure and Breach or Runout Assessment

10.5.5 (1) The manager must ensure the engineer of record develops a failure and breach or runout assessment for each TSF and dam that contains the following:

(a) an analysis of the failure modes and the expected results of each failure mode;

(b) potential dam failure scenarios;

(c) estimates of inundation or runout areas, if applicable;

(d) estimates of breach and arrival times, if applicable.

(2) The manager must ensure that the assessment under subsection (1) is reviewed by the engineer of record and updated prior to a material change to the design, construction, operation or downstream conditions that has affected or may affect the potential inundation or run out area.

(3) Repealed.

(4) Repealed.

Classification of TSFs and Dams

10.5.6 (1) The manager must ensure that, for each TSF and dam, the engineer of record

(a) determines the potential failure scenarios to be used to determine the consequences of the potential failure scenarios as described in Table 10-3, and

(b) reviews the failure and breach or runout assessment as set out in section 10.5.5 in making the determination under paragraph (a).

(2) The manager must ensure that the engineer of record documents the following for each TSF and dam:

(a) a determination, in accordance with Table 10-3, of the consequences of potential failure scenarios for the TSF or dam with input from other qualified professionals and persons with relevant areas of knowledge, as needed and appropriate;

(b) changes to the consequences of potential failure scenarios.

(3) The manager must ensure that, in determining the consequences of potential failure scenarios referred to in subsection (2) (a),

(a) reasonable efforts are made to engage with potentially affected First Nations, and

(b) local Indigenous knowledge received under paragraph (a) is considered.

Table 10-3
Consequences of Potential TSF or Dam Failure Scenarios

Potential Losses				
Consequence of Potential TSF or Dam Failure Scenarios	**Potential Loss of Life**	**Environment**	**Health, Social & Cultural**	**Infrastructure and Economics**
Low	none	Minimal short-term loss of environmental values. No expected impact on livestock or fauna drinking water. Limited area of impact and restoration feasible in short term.	Minimal effects and disruption of business and livelihood. No measurable effects on human health. No disruption of heritage, recreation, community or cultural assets.	Low economic loss: area contains limited infrastructure or services.
Significant	none	Limited loss or deterioration of environmental values. Potential contamination of livestock or fauna water supply. Moderate area of impact and restoration possible.	Limited effects and disruption of business and livelihood. No measurable effects on human health. Limited loss of regional heritage, recreation, community or cultural assets.	Moderate economic loss: losses to recreational facilities, seasonal workplaces and infrequently used transportation routes.
High	1 - 10	Significant loss or deterioration of critical environmental values. Potential contamination of livestock or fauna water supply. Potential area of impact 5 km^2 to 20 km^2. Restoration possible within a moderate time frame.	Many people affected by disruption of business, services or social dislocation. Significant loss of regional heritage, recreation, community or cultural assets. Potential for some short-term human health effects.	High economic loss: losses affecting infrastructure, public transportation, commercial facilities or employment. Moderate relocation costs and/or compensation to communities.

Potential Losses				
Consequence of Potential TSF or Dam Failure Scenarios	**Potential Loss of Life**	**Environment**	**Health, Social & Cultural**	**Infrastructure and Economics**
Very High	10 to 100	Major loss or deterioration of critical environmental values including rare and endangered species of high significance. Potential area of impact >20 km^2. Restoration or compensation possible but very difficult and requires a moderate to long time frame.	A high number of people affected by disruption of business, services or social dislocation for more than one year. Significant loss of national heritage, recreation, or community facilities or cultural assets. Significant long-term human health effects.	Very high economic loss: losses affecting important infrastructure, services (e.g., highway, industrial facilities or storage facilities for dangerous substances) or employment. High relocation costs and/or compensation to communities.
Extreme	> 100	Catastrophic loss of critical environmental values including rare and endangered species of high significance. Very large areas of potential impact. Restoration or compensation in kind impossible or requires a very long time.	A large number of people affected by disruption of business, services, or social dislocation for years. Significant national heritage or community facilities or cultural assets destroyed. Potential for Severe and/or long-term human health effects.	Extreme economic loss: losses affecting critical infrastructure or services (e.g., hospital, major industrial complex, major storage facilities for dangerous substances or employment. Very high relocation costs and/or compensation to communities and very high social readjustment costs.

Seismic and Flood Design Criteria

10.5.7 (1) The manager must ensure that the engineer of record designs each TSF or dam so it meets the minimum seismic and flood criteria set out in Table 10-4.

(2) The manager must ensure that, if a dam contains flowable tailings or water containing tailings, it meets the seismic and flood criteria set out in Table 10-4 for TSFs.

(3) The manager must ensure that the engineer of record designs the inflow design flood of each TSF or dam so that it addresses the following:
   (a) evaluation of scenarios of frequency, intensity and duration to identify controlling events;
   (b) consideration of rain on snow;
   (c) consideration of the effect of the seasons.

(4) The manager must ensure that the criteria respecting the environmental design flood is determined by the engineer of record.

(5) The manager must ensure that, when the criteria respecting the environmental design flood criteria is determined,
   (a) reasonable efforts are made to engage with affected First Nations, and
   (b) local Indigenous knowledge received under paragraph (a) is considered.

(6) Despite subsection (1), the manager may, for final closure of a TSF or dam classified as low consequence of potential failure as set out Table 10-3, apply to the chief permitting officer for an exemption to the final closure flood criteria, final closure seismic criteria, or both, as set out in Table 10-4.

(7) Subject to subsection (8), if, on the date this section comes into force, a TSF or dam exists and the manager does not meet a requirement as set out in subsections (1) to (5) of this section, with respect to the TSF or dam,
   (a) the manager is not required to meet the requirement with respect to the TSF or dam until November 1, 2026, and
   (b) despite its repeal, section 10.1.8 of the code, as it read immediately before its repeal, continues to apply to the TSF or dam in respect of the subject matter of the requirement referred to in paragraph (a) of this subsection, until the earlier of the following:
      (i) the manager meets the requirement;
      (ii) November 1, 2026.

(8) If, immediately before the date this section comes into force, an exception respecting section 10.1.8, as set out in section 10.1.19 (2) of the code, as it read immediately before its repeal, applied to a TSF or dam, the exception continues to apply to the TSF or dam until the earlier of the following:
   (a) section 10.2.10 (2) applies to the TSF or dam;
   (b) November 1, 2026.

Table 10-4
Minimum Seismic and Flood Criteria – Annual Exceedance Probability

<table>
<tr><th rowspan="2">Failure Consequence Classification</th><th colspan="3">Flood Criteria</th><th colspan="3">Seismic Criteria</th></tr>
<tr><th>TSFs</th><th>Dams</th><th>Final Closure</th><th>TSFs</th><th>Dams</th><th>Final Closure</th></tr>
<tr><td>Low</td><td rowspan="2">1/3rd Between 1/1,000 and PMF</td><td>1/200</td><td rowspan="5">PMF</td><td rowspan="2">1/2,475</td><td>1/200</td><td rowspan="5">1/10,000 or MCE</td></tr>
<tr><td>Significant</td><td>between 1/200 and 1/1,000</td><td>between 1/200 and 1/1,000</td></tr>
<tr><td>High</td><td colspan="2">1/3rd Between 1/1,000 and PMF</td><td colspan="2">1/2,475</td></tr>
<tr><td>Very High</td><td colspan="2">2/3 Between 1/1,000 and PMF</td><td colspan="2">1/2 Between 1/2,475 and 1/10,000 or MCE</td></tr>
<tr><td>Extreme</td><td colspan="2">PMF</td><td colspan="2">1/10,000 or MCE</td></tr>
</table>

Footnotes for Table 10-4

PMF means Probable Maximum Flood
MCE means Maximum Credible Earthquake

Design Slopes

10.5.8 (1) For a TSF or dam that has an overall downstream slope steeper than 2H:1V, the manager must submit justification by the engineer of record for the selected design slope and receive authorization by the chief permitting officer prior to construction.

(2) Subject to subsection (3), if, on the date this section comes into force, a TSF or dam exists and has an overall downstream slope steeper than 2H:1V,

(a) the manager is not required to comply with subsection (1) of this section with respect to the TSF or dam until November 1, 2026, and

(b) despite its repeal, section 10.1.9 of the code, as it read immediately before its repeal, continues to apply to the TSF or dam until the earlier of the following:

(i) the manager complies with subsection (1) of this section;

(ii) November 1, 2026.

(3) If, immediately before the date this section comes into force, an exception respecting section 10.1.9, as set out in section 10.1.19 (2) of the code, as it read immediately before its repeal, applied to a TSF or dam, the exception continues to apply to the TSF or dam until the earlier of the following:

(a) section 10.2.10 (2) applies to the TSF or dam;

(b) November 1, 2026.

Minimum Factors of Safety

10.5.9 (1) The manager must ensure that all TSFs and dams meet the criteria for minimum factors of safety set out in Table 10-5.

(2) If a TSF or dam has a calculated factor of safety that is less than the criteria for minimum factors of safety set out in Table 10-5 that must be met under subsection (1), the manager must submit justification by the engineer of record for the selected factor of safety and receive authorization by the chief permitting officer prior to construction.

(3) Subject to subsection (4), if, on the date this section comes into force, a TSF or dam exists,

(a) the manager is not required to comply with subsection (1) of this section with respect to the TSF or dam until November 1, 2026, and

(b) despite its repeal, section 10.1.10 of the code, as it read immediately before its repeal, continues to apply to the TSF or dam until the earlier of the following:

(i) the manager complies with subsection (1) of this section;

(ii) November 1, 2026.

(4) If, immediately before the date this section comes into force, an exception respecting section 10.1.10, as set out in section 10.1.19 (2) of the code, as it read immediately before its repeal, applied to a TSF or dam, the exception continues to apply to the TSF or dam until the earlier of the following:

(a) section 10.2.10 (2) applies to the TSF or dam;

(b) November 1, 2026.

Table 10-5
Minimum Factors of Safety for TSFs and Dams

Facility	Minimum Factors of Safety			
	Prior to storage of water and tailings	**Operations and Closure**	**Rapid Drawdown – upstream slope where applicable**	**Post Seismic**
TSF or dam	1.3	1.5	1.2 to 1.3	1.2

Plans for Underground Dams and Bulkheads

10.5.10 (1) The manager must ensure that issued for construction drawings, specifications, and quality assurance and quality control plans respecting structures for impounding water, restraining saturated material or confining air under pressure in an underground opening have been prepared by a professional engineer prior to construction.

(2) The manager must ensure that, prior to using an underground structure referred to in subsection (1), a professional engineer verifies the structure has been constructed in a manner consistent with the drawings, specifications and plans referred to in subsection (1) and is suitable for the intended use.

(3) Structures in subsection (1) required in the approved closure plan must be authorized by the chief permitting officer prior to construction.

(4) Subsections (1) to (3) do not apply in the case of
(a) a structure less than 1 m in height used solely for
(i) diverting the ordinary drainage, or
(ii) storing water for mining purposes, or
(b) ventilation bulkheads or regulators used solely for ordinary ventilation.

(5) Despite subsection (1), a temporary dam or bulkhead may be constructed in an underground mine during an emergency without meeting the requirements of subsection (1) but no person may be allowed to work in any part of the underground mine that could be affected by the construction or failure of the dam or bulkhead until a professional engineer has verified that the structure is suitable for the intended use.

TSFs and Dams – Operation and Closure

Management Systems

10.6.1 The manager of a mine with one or more TSFs or dams must
(a) develop and maintain a management system commensurate with the overall complexity of the TSFs and dams and include regular system audits, and
(b) ensure that local Indigenous knowledge received from a First Nation under this Part is not disclosed without prior written consent from the First Nation.

Dam Safety Review

10.6.2 (1) The manager must ensure that a dam safety review report for each TSF or dam is prepared by an independent professional engineer at the frequency set out in Table 10-6.

(2) The manager must ensure that the dam safety review report referred to in subsection (1) includes the following for each TSF or dam:
(a) a summary describing the TSF or dam and its components;
(b) identification of material changes to the infrastructure TSF or dam since the previous review, as applicable;
(c) a review of the consequences of potential failure scenarios;

(d) a review of the design, construction, operation and monitoring of the facility and assessment of its performance;
(e) a review of the design summary document and design criteria;
(f) the findings of the dam safety review prepared under this section including any recommended actions and associated timelines;
(g) a signed and sealed assurance statement by a professional engineer, in a form specified by the chief inspector.

(3) The manager must address each recommended action referred to in subsection (2) (f) by
(a) ensuring the engineer of record reviews each recommended action and either accepts it or proposes an alternate course of action, and
(b) completing each recommended action or alternate course of action, as determined by the engineer of record under paragraph (a), within timelines agreed to with the engineer of record.

(4) The manager must provide to the chief inspector the following documents:
(a) the dam safety review report referred to in subsection (1), by June 1 of the year following the year the report is required to be prepared;
(b) an annual summary of all the TSF and dam safety recommended actions referred to in subsection (3) (a), by March 31 of the following year.

(5) The manager must
(a) ensure that reasonable efforts are made to engage with each affected First Nation in order for each First Nation to identify if it wants to receive any of the documents set out in subsection (4), and
(b) provide to each First Nation a copy of each of the most recent documents identified by the First Nation under paragraph (a) of this subsection, in accordance with the timeframe applicable with respect to the chief inspector.

Table 10-6
Schedule of Dam Safety Review Periods

Facility	Duration Between Dam Safety Reviews based on consequences of potential failure as determined under Table 10-3		
	Category 1A, 1B and 2 dams	Low	Significant to Extreme
TSF	Not required	10 years	5 years
Dam		Not required	5 years

TSFs and Dams Register

10.6.3 (1) The manager must ensure that a register of all TSFs and dams, in a form specified by the chief inspector, is established and submitted to the chief inspector.

(2) The manager must ensure that the register is annually updated and provided to the chief inspector by March 31 of the following year.

(3) The manager must

(a) ensure that reasonable efforts are made to engage with each affected First Nation in order for each First Nation to identify if it wants to receive the register established under subsection (1) or updated under subsection (2), whichever is the most recent, and

(b) provide a copy of the most recent register to each First Nation that identifies under paragraph (a) of this subsection it wants to receive it, in accordance with the timeframe applicable with respect to the chief inspector.

Annual Facility Performance Report

10.6.4 (1) The manager must ensure that the engineer of record completes a facility performance report annually for each TSF or dam.

(2) The annual facility performance report must include the following for each TSF or dam:

(a) identification of whether the consequences of potential failure scenarios as described in Table 10-3 remain appropriate;

(b) a description of any material changes to the design, construction, operation and closure of each TSF and dam, their effect on the safety of the TSF or dam and whether they have been captured in the change register under section 10.6.9;

(c) a description of any updates to the design summary document and design criteria;

(d) a summary of construction and operation activities;

(e) a summary and analysis of the results of surveillance, instrumentation and monitoring;

(f) a review of potential TSF or dam failure modes during the review period;

(g) identification of whether the facility TSF or dam was operated during the review period according to the quantifiable performance objectives set out in section 10.5.3 (1) (c), and summarized and updated under section 10.5.4 and section 10.6.7 (6);

(h) incident reports;

(i) TSF or dam safety recommended actions, including prioritization rankings and timelines for completion.

(3) The manager must ensure that each TSF or dam safety recommended action included in the annual facility performance report under

subsection (2) (i) is implemented within the timeline recommended under that provision unless the engineer of record agrees, in writing, to an alternate course of action or timeline, in which case the manager must ensure the alternate course of action or timeline is implemented.

(4) The manager must ensure that the engineer of record provides a signed and sealed annual facility performance report assurance statement, in a form specified by the chief inspector.

(5) The manager must annually provide to the chief inspector, by March 31 of the following year,
- (a) the facility performance report referred to in subsection (1), and
- (b) a summary of all the TSF and dam safety recommended actions, referred to in subsection (2) (i).

(6) The manager must
- (a) ensure that reasonable efforts are made to engage with each affected First Nation in order for each First Nation to identify if it wants to receive any of the documents referred to in subsection (5), and
- (b) provide to each First Nation a copy of each of the most recent documents identified by the First Nation under paragraph (a) of this subsection, in accordance with the timeframe applicable with respect to the chief inspector.

Construction

10.6.5 (1) The manager must ensure that the following documents are prepared by the engineer of record for each TSF or dam:
- (a) issued for construction drawings;
- (b) issued for construction specifications;
- (c) a summary of milestones and key timelines associated with constructing the TSF or dam;
- (d) quality assurance and quality control requirements that meet the issued for construction specifications referred to in paragraph (b).

(2) The manager must ensure that the engineer of record has verified, in a form specified by the chief inspector, that the TSF or dam is ready to receive tailings or water prior to use and submitted the signed and sealed verification to the chief inspector prior to the TSF or dam receiving the tailings or water.

(3) The manager must ensure that the engineer of record completes a construction records report when construction begins on a TSF or dam, and annually until construction is completed, and includes the following respecting the construction of the TSF or dam:
- (a) geotechnical foundation conditions;
- (b) as constructed representative cross-sections;
- (c) quality assurance and quality control data;

(d) interpretation of the data;
(e) a description of any unforeseen deviations or material changes from subsection (1) (a) or (b);
(f) installed instrumentations.

(4) The manager must ensure that the construction records report includes a signed and sealed assurance statement, in a form specified by the chief inspector, from the engineer of record that the facility substantially complies in all material respects with the original design intent and that the TSF or dam is suitable for use.

(5) Unless otherwise stated in the permit, the manager must provide to the chief inspector
(a) the documents set out in subsection (1), prior to each stage of construction, and
(b) the construction records report as referred to in subsection (3), by June 1 of the year following the year the report is required to be completed.

(6) The manager must
(a) ensure that reasonable efforts are made to engage with each affected First Nation in order for each First Nation to identify if it wants to receive any of the documents described in subsections (5) (a) and (b), and
(b) provide to each First Nation a copy of each of the most recent documents identified by the First Nation under paragraph (a) of this subsection, in accordance with the timeframe applicable with respect to the chief inspector.

Operations, Maintenance and Surveillance (OMS) Manual

10.6.6 (1) The manager must ensure an Operations, Maintenance and Surveillance (OMS) Manual is prepared by one or more qualified persons for each TSF or dam respecting the operations and closure of the TSF or dam.

(2) The manager must ensure that the OMS Manual is reviewed by the engineer of record, unless it is prepared by the engineer of record.

(3) Prior to implementation of the OMS Manual, the manager must
(a) ensure that recommendations from the engineer of record, after a review under subsection (2), are addressed, and
(b) review and approve the OMS manual.

(4) The manager must ensure that the OMS Manual is implemented prior to the initial filling of the TSF or dam with tailings or water.

(5) The OMS Manual must include the following:
(a) a description of roles, responsibilities and training requirements;
(b) consequences of potential failure scenarios and key design requirements of the TSFs and dams;

(c) an instrumentation, monitoring and surveillance plan;
(d) the quantifiable performance objectives set out in section 10.5.3 (1) (c), and summarized and updated under section 10.5.4 and section 10.6.7 (6);
(e) trigger action response plans in cases of escalating changes of dam safety conditions set out in section 10.6.7 (6);
(f) maintenance and testing requirements for key equipment for safe operation of the TSF or dam.

(6) The manager must ensure that all employees and contractors involved in the construction or operation of a TSF or dam are trained and qualified, based on the OMS Manual, prior to commencing work at the TSF or dam.

(7) The manager must ensure that the OMS Manual is
(a) reviewed annually by the TSF qualified person or dam qualified person, as applicable, and the engineer of record, and
(b) updated when the engineer of record considers it appropriate.

(8) The manager must ensure that, whenever the OMS Manual is updated, employees and contractors are provided with additional training, as appropriate.

(9) The manager must keep the OMS Manual on site and make it available to an inspector on request.

(10) Repealed.

Water Management

10.6.7 (1) The manager must ensure that a qualified professional develops an overall site water balance and overall water management plan for the mine.

(2) The manager must ensure that a qualified professional
(a) reconciles the overall site water balance annually, and
(b) updates the overall water management plan when there are material changes.

(3) The manager must ensure that, when the overall site water balance and overall water management plan is developed under subsection (1) or the overall site water balance is reconciled under subsection (2),
(a) reasonable efforts are made to engage with affected First Nations, and
(b) local Indigenous knowledge received under paragraph (a) is considered.

(4) The manager must ensure that a qualified professional develops a water balance and water management plan for each TSF and dam that is
(a) integrated with the overall site water balance and overall water management plan referred to in subsection (1), and

(b) based on the design summary document as set out in section 10.5.4.

(5) Respecting the water balance and water management plan referred to in subsection (4), the manager must ensure that a qualified professional
   (a) reconciles the water balance annually, and
   (b) updates the water management plan when there are material changes.

(6) The manager must ensure the engineer of record
   (a) develops quantifiable performance objectives, and trigger action response plans in cases of escalating changes of dam safety conditions, that are informed by the water balance and water management plan for the TSF or dam, and
   (b) reviews and updates the quantifiable performance objectives, and trigger action response plans, as set out in paragraph (a) when the engineer of record considers it appropriate.

(7) The manager must ensure that
   (a) the quantity of water predicted by the water balance can be safely stored, and
   (b) surplus water balances are reduced in accordance with a plan approved by the engineer of record.

(8) The manager must ensure that the water balance and water management plans for each TSF or dam referred to in subsection (4) include the following:
   (a) water usage, water sources and discharges from the mine;
   (b) layout of all water management infrastructure;
   (c) location of monitors and methods;
   (d) water balance schematics;
   (e) climate conditions and hydrology;
   (f) groundwater and surface water interactions;
   (g) water balance projections.

(9) The manager must immediately notify the chief inspector and affected communities and First Nations if any water discharged without a permit occurs or is necessary.

(10) The manager must provide to the chief inspector
   (a) the annual reconciliation of the overall site water balance referred to in subsection (2), by March 31 of the year following the year it is reconciled,
   (b) the updated overall water management plan referred to in subsection (2), by March 31 of the year following the year it is updated, and
   (c) the updated water management plan referred to in subsection (5), by March 31 of the year following the year it is updated.

(11) The manager must
   (a) ensure that reasonable efforts are made to engage with each affected First Nation in order for each First Nation to identify if it wants to receive any of the documents referred to in subsection (10), and
   (b) provide to each First Nation a copy of each of the most recent documents identified by the First Nation under paragraph (a) of this subsection, in accordance with the timeframe applicable with respect to the chief inspector.

(12) Repealed.

(13) Repealed.

(14) Repealed.

Risk Assessment

10.6.8 (1) Prior to the initial filling of the TSF or dam with tailings or water, the manager must ensure that a risk assessment is prepared by a qualified professional, with experience commensurate with the complexity of the TSF or dam, that documents the likelihood of potential failure scenarios and the consequences of potential failure scenarios.

(2) The manager must ensure that the engineer of record annually reviews the risk assessment to ensure that the following, at a minimum, are current and appropriate to manage risks:
   (a) the quantifiable performance objectives set out in section 10.5.3 (1) (c), and summarized and updated under section 10.5.4, and section 10.6.7 (6);
   (b) the operating and monitoring requirements set out in the Operations Maintenance and Surveillance Manual prepared under section 10.6.6.

Change Register

10.6.9 (1) The manager must develop and maintain a change register, in consultation with the engineer of record, to track material changes to the design, construction, operation and closure of each TSF and dam.

(2) The manager must ensure that the engineer of record acknowledges and addresses all material changes entered into the change register.

(3) The manager must ensure that the change register is made available to an inspector on request.

Emergency Preparedness and Response Plan

10.6.10 (1) The manager must develop an Emergency Preparedness and Response Plan (EPRP) for potential TSF or dam failures that is included in the Mine Emergency Response Plan (MERP) required

under section 3.7.1 (2) (c) and that contains the following minimum components:

(a) one or more maps showing potential impact zones and potentially affected infrastructure;
(b) emergency escalation levels detailing escalation triggers;
(c) stakeholders, potentially affected First Nations and community warnings and notifications;
(d) emergency responses and procedures including evacuation of mine personnel;
(e) roles, responsibilities and contact information of key personnel.

(2) The manager must ensure that testing of the Emergency Preparedness and Response Plan is conducted in accordance with the following:
(a) functional tests of parts of the plan are conducted annually;
(b) a functional test of the full plan is conducted, at a minimum, every 3 years;
(c) reasonable efforts are made to include potentially affected communities and potentially affected First Nations in the testing under paragraphs (a) and (b).

(3) The manager must ensure that
(a) all identified issues or recommendations as a result of the testing under subsection (2) (a) and (b) are addressed, and
(b) the Emergency Preparedness Response Plan is kept up to date.

(4) Repealed.

Climate Change

10.6.11 (1) The manager must engage qualified professionals to assess the hydrological and climate conditions, and other relevant conditions associated with climate change, for each TSF and dam at least every 5 years.

(2) The climate change assessment described in subsection (1) must be reviewed by the engineer of record for the TSF or dam and incorporated into the design summary document as set out in section 10.5.4.

Closure Design Report for a TSF or Dam

10.6.12 (1) The manager must ensure that, for each TSF or dam, the engineer of record develops a closure design report that shows how closure of the TSF or dam is feasible.

(2) Despite subsection (1), if, on the date this subsection comes into force, a closure design report for each TSF and dam as described in subsection (1) has not already been submitted to the chief permitting officer under section 10.2.2 (i), the manager must, within 3 years after the date this subsection comes into force,

(a) ensure the engineer of record develops a closure design report for each TSF or dam in accordance with subsection (1), and
(b) provide each closure design report referred to in paragraph (a) of this subsection to the chief inspector.

(3) The manager must ensure that each closure design report referred to in subsection (1) or (2) (a) is
(a) updated by the engineer of record at least every 5 years, and
(b) provided to the chief inspector by March 31 of the year following the update.

(4) No less than 3 years prior to a planned closure of a TSF or dam, the manager must
(a) ensure that the engineer of record updates the closure design report for the TSF or dam that shows in detail
(i) how the TSF or dam will achieve closure, and
(ii) the schedule for implementation, and
(b) provide the closure design report to the chief inspector by March 31 of the year following the update.

(5) If a closure design report is provided to the chief inspector under subsection (4) (b) and, subsequently, the permit respecting the TSF or dam is extended so that the planned closure date is 5 years or longer from the date the updated report is provided to the chief inspector, the manager may update and provide the closure design report of the TSF or dam in accordance with subsection (3) (a) and (b), until such time as subsection (4) applies with respect to the TSF or dam.

(6) A closure design report developed or updated under this section must
(a) address the following:
(i) physical stability for potential long-term changes to slope stability, floods and water erosion, and other natural or mine-induced hazards;
(ii) long-term prevention, mitigation and management of metal leaching and acid rock drainage for the offsite release of mine-affected groundwater and surface water;
(iii) ecological and landform aspects that influence closure;
(iv) land and water use objectives,
(b) include the following:
(i) a design of permanent spillways and other necessary civil works;
(ii) a cost estimate and schedule for implementation;
(iii) a long-term monitoring plan, and
(c) demonstrate how the TSF or dam will meet the criteria for final closure as set out in Table 10-4, subject to section 10.5.7 (6).

(7) The manager must ensure that, in developing the land and water use objectives referred to in subsection (6) (a) (iv),
   (a) reasonable efforts are made to engage with local communities and affected First Nations, and
   (b) local Indigenous knowledge received under paragraph (a) is considered.

(8) The manager must
   (a) ensure that reasonable efforts are made to engage with each affected First Nation in order for each First Nation to identify if it wants to receive each closure design report developed or updated under this section, whichever is the most recent, and
   (b) provide a copy of the most recent report to each First Nation that identifies under paragraph (a) of this subsection it wants to receive it, in accordance with the timeframe applicable with respect to the chief inspector.

(9) If, on the date this section comes into force, the planned closure of a TSF or dam is less than 3 years, or a TSF or dam is inactive, the manager must,
   (a) within 3 years after the date this section comes into force,
      (i) ensure that the engineer of record updates the closure design report for the TSF or dam that shows in detail
         (A) how the TSF or dam will achieve closure, and
         (B) the schedule for implementation, and
      (ii) provide the closure design report to the chief inspector by March 31 of the year following the update, and
   (b) in meeting the requirements of paragraph (a) of this subsection, comply with subsections (6), (7) and (8).

Closed Tailings Storage Facility or Dams

10.6.13 (1) A tailings storage facility that has not been operated for a period of 12 or more months may be declared closed by the chief inspector.

(2) If the chief inspector declares a TSF closed under subsection (1), the manager
   (a) must not reactivate the TSF unless the chief permitting officer grants another permit that allows the reactivation, and
   (b) must not deposit tailings into the TSF unless the TSF has been reactivated in accordance with paragraph (a).

(3) The manager of a TSF or dam that has achieved final closure and approval of the engineer of record may apply to the chief permitting officer for the release of permit obligations under the *Mines Act*.

(4) If the manager of a TSF or dam has completed substantial work towards final closure of the TSF or dam but has not been granted release of permit obligations as referred to in subsection (3), the